Coordinating language selection in speech planning scopes: phrasal lexical costs vs. local phonetic costs in bilingual sentence production

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Abstract #91

Background

Incremental speech production models propose multiple processing stages (Levelt et al., 1999)

- Lexical level: phrasal planning
- Phonological/phonetic level: word-by-word planning

Default language & switch cost

- In connected speech, bilinguals adopt a **default language** that guides lexical choice and syntactic structure;
- Violation of default language selection elicited robust switch costs (Gollan & Goldrick, 2018);
- Switch costs have been typically measured by reaction time/response latency, hard to measure in connected speech where the planning of upcoming words occurs alongside the production of current ones
- This necessitates other measurements to examine how bilinguals pay switch costs in connected speech

Research question:

- At what point in a sentence, do bilinguals pay the costs of the violation of default language selection, at lexical and phonetic levels?
  - *Lexical level*: speech onset latency and word duration
  - *Phonetic level*: vocal effort
- Do lexical-level costs align with phonetic-level costs temporally?

Methods

Participants

- 30 Spanish-English bilinguals; Dominant in English

Design & Procedure

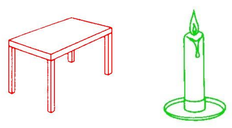
- 2 (switch vs. stay) × 2 (simple-initial vs. complex-initial)

Training Phase 1

- Name black & white pictures in English

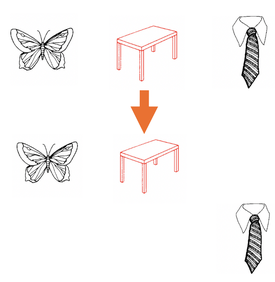
Training Phase 2 (to elicit accessibility driven-switches)

- Name Red vs.Green pictures in English vs. Spanish
  - Color-picture & Color-language correspondences are counterbalanced

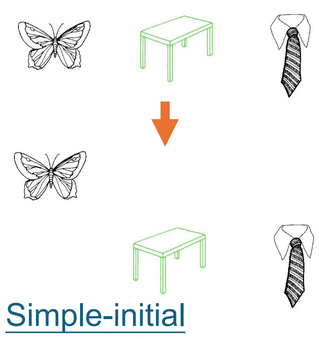


Display Description

- Describe moving pictures in certain structure



Complex-initial
[The butterfly and the mesa] [moved above] [the tie].



Simple-initial
[The butterfly] [moved above] [the table and the tie].

Acoustic correlates of vocal effort

- The three acoustic measures were extracted from **the vowel of the stressed syllable** of each word in the sentence.

RMS Energy	Higher	→	Greater loudness
H1*–H2* (Amplitude diff between 1st and 2nd harmonics)	Lower	→	Greater glottal constriction
HNR (Harmonic-to-noise ratio)	Higher	→	More harmonic voicing

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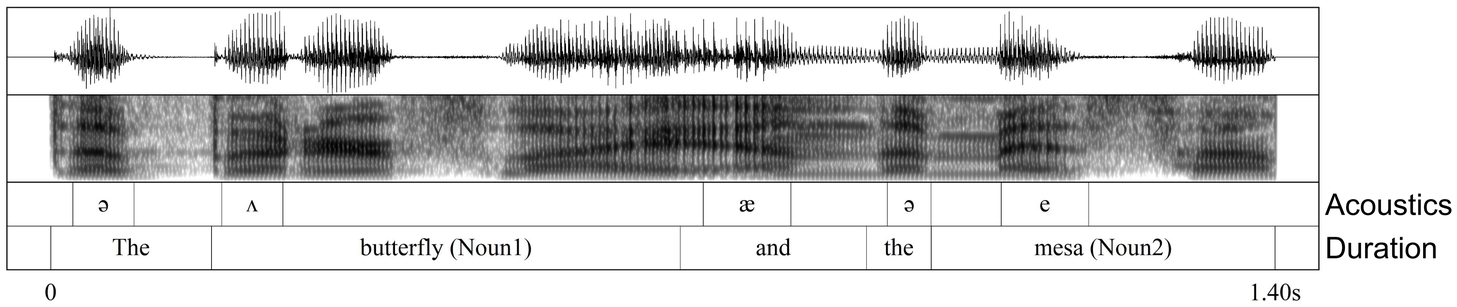
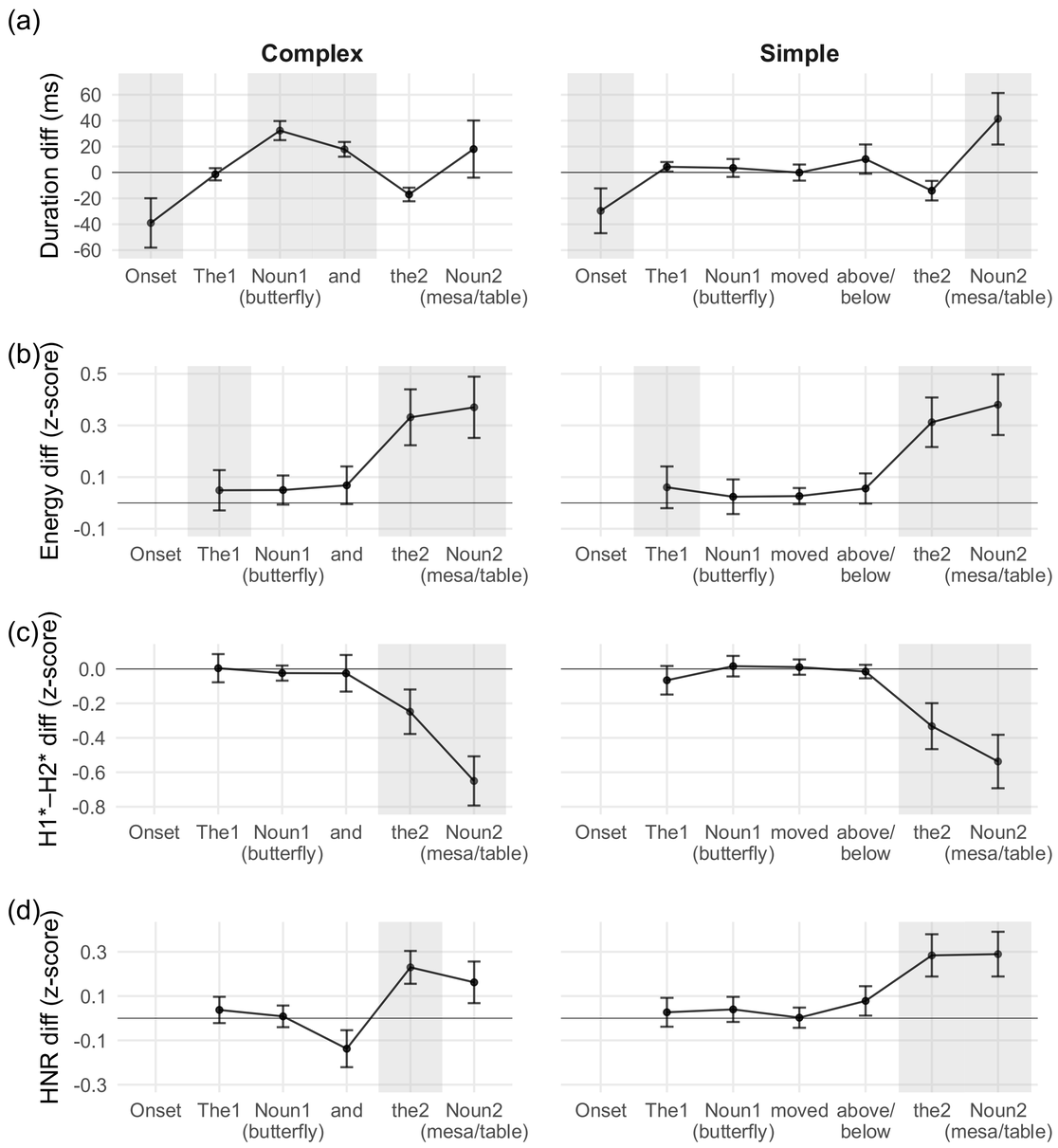
Results

- Acoustics measures were extracted from the vowel of the stressed syllable.
- Duration was calculated for the entire word.





Compared with Stay condition, Switch condition has:

Shorter speech onset latency

Increased duration: Complex: **Noun1 & and** Simple: **Noun2**

Larger loudness

Greater glottal constriction

More harmonic voicing

Increased vocal efforts in **Noun2** & **the2**

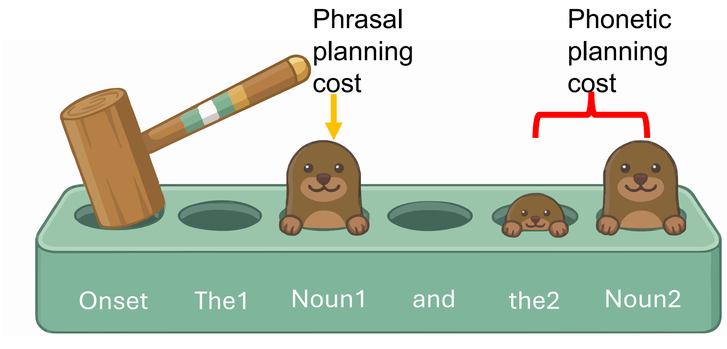
Conclusions

Temporal asynchrony between lexical and phonetic switch cost:

- Lexical level: syntactic structure–dependent; slowing occurs at the first noun in the phrase containing language switching, consistent with phrasal planning
- Phonetic level: structure–independent; effects occur locally at the switch word and preceding determiner.

Redistribution of switch cost: A "Whack-a-mole" pattern

- Switch costs of default language violations are paid at different time points at different processing stage.
- It is important to use multidimensional measures to understand how language control mechanisms are applied in real-time at different levels of the speech production hierarchy



Scan for supplementary materials and references